AI ASSISTANT CODING
ASSIGNMENT-7.2

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Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs

Task Description

Task 1 – Runtime Error Due to Invalid Input Type

• A Python program accepts user input and performs arithmetic operations. However, the program throws a runtime error because the input is treated as a string instead of a numeric type.

Example (Buggy Code):

```
num = input("Enter a number: ")

result = num + 10

print(result)
```

• **Task:**

Use AI tools to identify the cause of the runtime error and modify the program so it executes correctly.

```python
"""Prompt - “Identify the runtime error in the following Python program
where user input is used in an arithmetic operation. Explain why the
error occurs and modify the code so that the program executes correctly
without errors."""

# Buggy code:
num = input("Enter a number: ")
res = num + 10
print(res)
```

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Lab_Ass_7.2.py"
Enter a number: 10
Traceback (most recent call last):
  File "c:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING\Lab_Ass_7.2.py", line 8, in <module>
    res = num + 10
          ~~~~^~~~
TypeError: can only concatenate str (not "int") to str
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

```python
"""Fixed code"""

# Take input from the user
# input() always returns the value as a string
num = input("Enter a number: ")

# Convert the string input to an integer
# This is required to perform arithmetic operations
num = int(num)

# Perform arithmetic operation (now both are integers)
result = num + 10

# Print the final result
print(result)
```

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Lab_Ass_7.2.py"
Enter a number: 110
120
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

Task Description:

Task 2 – Incorrect Function Return Value

A function is designed to calculate the square of a number, but it does not return the computed result properly.

Example (Buggy Code):

def square(n):

result = n * n

Task:

Use AI assistance to analyze the function and ensure the correct value is returned

```python
"""
Prompt - Analyze the following Python function that is intended to calculate the square of a number
but does not return the result correctly. Identify the issue in the function and
modify it so that it returns the correct output
"""
# Buggy code:
def square(n):
    # This line correctly calculates the square of the number
    result = n * n
    # There is no return statement in this function
    # Without 'return', Python automatically returns N

print(square(3))  # This will print 'None' instead of '9'
```

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Task_2.py"
None
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

```python
"""Fixed code"""
def square(n):
    # Calculate the square of the given number
    result = n * n

    # Return the calculated value to the caller
    # This ensures the function outputs the correct result
    return result
print(square(3))
```

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Task_2.py"
None
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Task_2.py"
9
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

Task Description

Task 3 – Index Error in List Traversal

A Python program iterates over a list using incorrect index limits, causing an IndexError.

Example (Buggy Code):

numbers = [10, 20, 30]

for i in range(0, len(numbers)+1):

print(numbers[i])

Task:

Use AI to identify the incorrect loop boundary and correct the iteration

logic.

```
"""
Prompt - Analyze the following Python program that raises an IndexError while iterating over a list.
Identify the incorrect loop boundary, explain why the error occurs, and modify the code to prevent
out-of-range list access."""
# Buggy code:
numbers = [10, 20, 30]

# Loop goes one step beyond the valid index range
for i in range(0, len(numbers)+1):
    print(numbers[i])
```

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Task_3.py"
10
20
30
Traceback (most recent call last):
  File "c:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING\Task_3.py", line 10, in <module>
    print(numbers[i])
          ~~~~~~~^^^
IndexError: list index out of range
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

```
"""Fixed code"""
numbers = [10, 20, 30]

# Loop runs from index 0 to len(numbers) - 1
for i in range(0, len(numbers)):
    print(numbers[i])
```

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Task_3.py"
10
20
30
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

Task Description

Task 4 – Uninitialized Variable Usage

A program uses a variable in a calculation before assigning it any value.

Example (Buggy Code):

if True:

pass

print(total)

Task:

Use AI tools to detect the uninitialized variable and correct the program.

Task_4.py

```python
"""Prompt - Analyze the following Python program that uses a variable before it is initialized.
Identify the uninitialized variable, explain why the error occurs, and modify the code so that the
variable is properly initialized before being used."""
# Buggy code:
if True:
    # pass is a placeholder statement
    pass

# 'total' is used here without being defined earlier
# Python raises NameError
print(total)
```

PROBLEMS 1 DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CO
DING/Task_4.py"
Traceback (most recent call last):
  File "c:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING\Task_4.py", line 11, in <module>
    print(total)
          ^^^^^
NameError: name 'total' is not defined
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

Task_4.py > ...

```python
"""Fixed code"""
# Initialize the variable before using it
total = 4

# The if block executes, but pass does nothing
if True:
    pass

# Now total is defined, so this works safely
print(total)
```

PROBLEMS DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CO
DING/Task_4.py"
4
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

Task Description

Task 5 – Logical Error in Student Grading System

A grading program assigns incorrect grades due to improper conditional logic.

Example (Buggy Code):

marks = 85

if marks >= 90:

grade = "A"

elif marks >= 80:

grade = "C"

else:

grade = "B"

print(grade)

Task:

Use AI to analyze the grading conditions and correct the logical flow.

Task_5.py > ...

```python
"""
Prompt - Analyze the following Python grading program that assigns incorrect grades due to logical
errors in conditional statements. Identify the issue in the grading logic, explain why the output
is incorrect, and correct the conditional flow so that grades are assigned accurately.
"""
# Buggy code:
marks = 85

if marks >= 90:
    grade = "A"
elif marks >= 80:
    grade = "C"
else:
    grade = "B"

print(grade)
```

PROBLEMS DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Task_5.py"
C
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```

Task_5.py > ...

```python
"""Fixed code"""
marks = 85

# Check for highest grade first
if marks >= 90:
    grade = "A"

# Check for the next lower grade range
elif marks >= 80:
    grade = "B"

# Remaining marks fall into the lowest grade
else:
    grade = "C"

print(grade)
```

PROBLEMS DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hp/OneDrive/Desktop/AI ASSISTANT CODING/Task_5.py"
B
PS C:\Users\hp\OneDrive\Desktop\AI ASSISTANT CODING>
```